# Further Integration Techniques

**ANSWER KEY**

## Section 1: Integration by Parts (Repeated & Cyclic)

**1** Evaluate $\int x^2 \cos x \, dx$ using integration by parts.

**a.** $x^2 \sin x + 2x \cos x - 2 \sin x + C$ ✓

**b.** $x^2 \sin x - 2x \cos x + 2 \sin x + C$

**c.** $-x^2 \sin x + 2x \cos x + C$

**d.** $\frac{1}{3} x^3 \sin x + C$

Tabular: $x^2 \to 2x \to 2 \to 0$; $\cos x \to \sin x \to -\cos x \to -\sin x$. Alternating: $+x^2 \sin x - 2x(-\cos x) + 2(-\sin x) = x^2 \sin x + 2x \cos x - 2 \sin x + C$.

**2** Evaluate the cyclic integral $I = \int e^x \sin x \, dx$.

**a.** $\frac{1}{2} e^x (\sin x - \cos x) + C$ ✓

**b.** $\frac{1}{2} e^x (\sin x + \cos x) + C$

**c.** $e^x (\sin x - \cos x) + C$

**d.** $-e^x \cos x + C$

Integrate by parts twice: $I = e^x \sin x - \int e^x \cos x \, dx = e^x \sin x - [e^x \cos x - \int e^x (-\sin x) dx] = e^x (\sin x - \cos x) - I \Rightarrow 2I = e^x (\sin x - \cos x) \Rightarrow$ .

**3** Evaluate $\int_1^e \ln x \, dx$.

**a.** $1$ ✓

**b.** $e - 1$

**c.** $e$

**d.** $0$

$\int \ln x \, dx = x \ln x - x$. Evaluating from 1 to $e$: $[e \ln e - e] - [1 \ln 1 - 1] = [e - e] - [0 - 1] = 0 - (-1) = 1$.

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## Section 2: Trigonometric Substitutions

**4** Which trigonometric substitution is best suited to evaluate $\int \frac{1}{\sqrt{16-x^2}}dx$?

**a.** $x = 4\sin\theta$ ✓

**b.** $x = 4\tan\theta$

**c.** $x = 4\sec\theta$

**d.** $x = 16\sin\theta$

For forms $\sqrt{a^2-x^2}$, the substitution $x = a\sin\theta$ simplifies the radical via $16 - 16\sin^2\theta = 16\cos^2\theta$.

**5** Evaluate $\int \frac{1}{x^2+16}dx$.

**a.** $\frac{1}{4}\arctan\left(\frac{x}{4}\right) + C$ ✓

**b.** $\arctan\left(\frac{x}{4}\right) + C$

**c.** $\frac{1}{16}\arctan\left(\frac{x}{16}\right) + C$

**d.** $\frac{1}{4}\ln|x^2+16| + C$

Standard formula $\int \frac{1}{x^2+a^2}dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$. With $a = 4$: $\frac{1}{4}\arctan\left(\frac{x}{4}\right) + C$.

**6** Which substitution evaluates $\int \sqrt{x^2-9}\,dx$?

**a.** $x = 3\sec\theta$ ✓

**b.** $x = 3\sin\theta$

**c.** $x = 3\tan\theta$

**d.** $x = 9\sec\theta$

For $\sqrt{x^2-a^2}$, use $x = a\sec\theta$, since $x^2 - 9 = 9(\sec^2\theta - 1) = 9\tan^2\theta$.

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## Section 3: Integrals of Powers of Trigonometric Functions

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**7** Evaluate $\int \sin^3 x dx$.

**a.** $-\cos x + \frac{1}{3}\cos^3 x + C$ ✓

**b.** $\cos x - \frac{1}{3}\cos^3 x + C$

**c.** $\frac{1}{4}\sin^4 x + C$

**d.** $-\frac{1}{3}\cos^3 x + C$

Rewrite $\sin^3 x = (1 - \cos^2 x)\sin x$. Let $u = \cos x$, $du = -\sin x dx$:
$\int -(1 - u^2)du = -u + \frac{u^3}{3} + C = -\cos x + \frac{1}{3}\cos^3 x + C$.

**8** Evaluate $\int \cos^2 x dx$.

**a.** $\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$ ✓

**b.** $\frac{1}{2}x - \frac{1}{4}\sin(2x) + C$

**c.** $\frac{1}{3}\cos^3 x + C$

**d.** $\frac{1}{2}x + \sin(2x) + C$

Use double angle identity $\cos^2 x = \frac{1 + \cos(2x)}{2}$: $\int \frac{1 + \cos(2x)}{2} dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$.

**9** Evaluate $\int \tan^2 x dx$.

**a.** $\tan x - x + C$ ✓

**b.** $\sec^2 x + C$

**c.** $\tan x + x + C$

**d.** $\frac{1}{3}\tan^3 x + C$

Identity: $\tan^2 x = \sec^2 x - 1$. Integrating: $\int (\sec^2 x - 1)dx = \tan x - x + C$.

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## Section 4: Integration by Substitution (Advanced Forms)

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**10** Evaluate $\int \frac{x}{\sqrt{x+1}}dx$ using substitution $u = x + 1$.

**a.** $\frac{2}{3}(x+1)^{3/2} - 2(x+1)^{1/2} + C$ ✓

**b.** $\frac{2}{3}(x+1)^{3/2} + 2(x+1)^{1/2} + C$

**c.** $(x+1)^{3/2} - (x+1)^{1/2} + C$

**d.** $\frac{1}{2}(x+1)^2 + C$

With $u = x + 1$, $x = u - 1$ and $dx = du$:
$\int \frac{u-1}{u^{1/2}}du = \int (u^{1/2} - u^{-1/2})du = \frac{2}{3}u^{3/2} - 2u^{1/2} + C = \frac{2}{3}(x+1)^{3/2} - 2(x+1)^{1/2} + C.$

**11** Evaluate $\int_0^1 \frac{e^x}{1+e^x}dx$.

**a.** $\ln\left(\frac{1+e}{2}\right)$ ✓

**b.** $\ln(1+e)$

**c.** $e - 1$

**d.** $\frac{e}{2}$

Let $u = 1 + e^x$, $du = e^x dx$. At $x = 0$, $u = 2$; at $x = 1$, $u = 1 + e$.
$\int_2^{1+e} \frac{1}{u}du = [\ln|u|]_2^{1+e} = \ln(1+e) - \ln 2 = \ln\left(\frac{1+e}{2}\right).$

**12** Evaluate $\int \frac{1}{x\ln x}dx$ for $x > 1$.

**a.** $\ln|\ln x| + C$ ✓

**b.** $\frac{1}{2}(\ln x)^2 + C$

**c.** $\frac{1}{\ln x} + C$

**d.** $x\ln x - x + C$

Let $u = \ln x$, $du = \frac{1}{x}dx$. $\int \frac{1}{u}du = \ln|u| + C = \ln|\ln x| + C.$

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